WEEK 4 CAPSTONE / DEMO DAY

Netixsol AI Client Intake & Project Scoping Agent

A retrieval-grounded assistant that answers prospective client questions and guides them toward a structured project enquiry.

THE PROBLEM

Prospective clients may not know which services fit their needs or what information is needed to scope a software project.

THE SOLUTION

A Gemini-powered RAG agent retrieves Netixsol knowledge, asks follow-up questions, and prepares an enquiry preview.

THE AUDIENCE

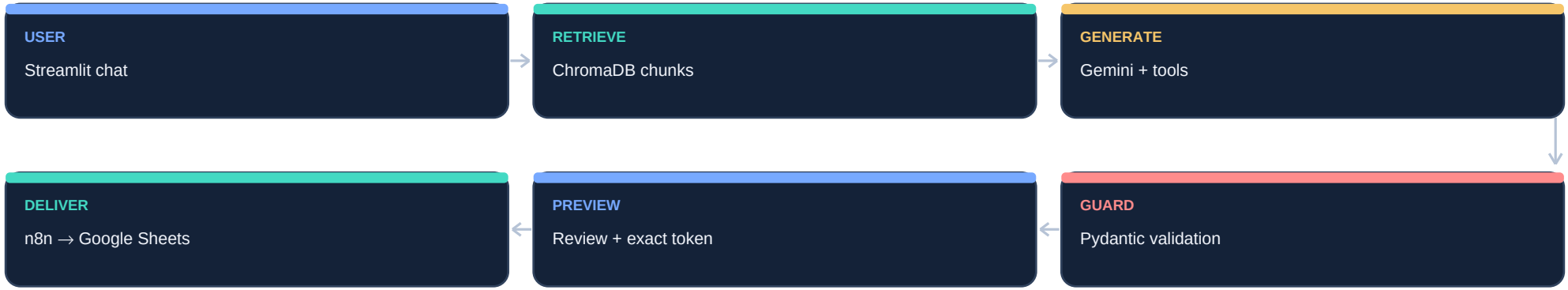
Businesses exploring AI agents, blockchain and DeFi development, or custom product engineering.

Core value: useful project discovery grounded in approved company information—with human review before an enquiry is sent.

02 / HOW IT WORKS

Architecture & user journey

The application connects retrieval, Gemini tool use, validation, and a human-controlled handoff.



Human review and confirmation gate the final handoff.

RETRIEVAL

ChromaDB supplies relevant company chunks. Answers display source links for inspection.

AGENT + VALIDATION

Gemini responds and gathers enquiry details. Pydantic validates structured fields and consent.

CONTROLLED HANDOFF

The user reviews a preview and enters its exact token. The configured n8n workflow can deliver the enquiry to Google Sheets.

03 / LIVE DEMO

A short, realistic demo

Show the experience from a grounded answer to a safe enquiry preview. Do not submit a real enquiry during the presentation.

01	**Ask about services**	“What services does Netixsol offer?” Show the answer and source links.
02	**Explore a project**	“Can you help scope a decentralized exchange with wallet integration?” Show follow-up questions.
03	**Test an unknown**	“How much does a mobile application cost?” Show that the agent does not invent a price.
04	**Demonstrate guardrails**	Ask for confidential client details. Show the refusal and explain the privacy boundary.
05	**Preview, do not send**	Show a fictional enquiry preview. Explain exact-token confirmation; leave it unsubmitted.

DEMO SAFETY NOTE

Use fictional details only. Keep the real confirmation token private and do not trigger a live webhook during rehearsal or Demo Day.

04 / EVALUATION

Measured against 10 test questions

The final evaluation records groundedness, behaviour checks, finish reasons, cited URLs, and retrieved evidence.

20 / 20	10 / 10	10
Groundedness points recorded	Behaviour checks recorded as pass	Evaluation questions

WHAT WAS TESTED

Service information; DEX and wallet scoping; project process; unknown pricing and delivery guarantees; website requirements; and confidential client information.

ITERATION

Four questions (3, 5, 6, and 7) were retested after prompt improvements. Before/after records are preserved in the evaluation files.

Q&A evidence: `eval_final.csv`, `eval_before_fix.csv`, and `eval_retrieval_evidence.json`. Scores are manually recorded; explain them using the underlying answers and retrieved chunks.

05 / TRUST & SAFETY

Guardrails before action

The agent assists with discovery; it must not make unsupported promises or silently submit client information.

GROUNDED ANSWERS

Use retrieved Netixsol information and show source links. If pricing or delivery guarantees are not specified, say so.

PRIVACY

Do not disclose confidential client information. Keep API keys and webhook credentials in local environment configuration.

VALIDATION

Validate enquiry fields and consent before producing a structured preview.

HUMAN APPROVAL

Require review of the preview and exact confirmation token before the handoff step.

TEST EVIDENCE

The offline self-test passed checks for exact-preview confirmation, duplicate prevention, changed-preview token invalidation, injection, email, category, and consent. Its webhook was mocked. A separate two-tab test showed that a token from one tab did not confirm the other tab’s preview.

06 / REFLECTION & NEXT STEPS

Prototype today. Stronger retrieval tomorrow.

The current knowledge base is small; success on this corpus does not guarantee performance at larger scale.

CURRENT LIMITATIONS

- Small knowledge base (five chunks in the initial retrieval experiment)
- Retrieved four of those five chunks in that experiment
- Retrieval quality needs testing on a broader corpus
- Persistent and concurrent-session behaviour deserves further hardening

NEXT IMPROVEMENTS

- Expand and clean the source corpus (e.g., 200 pages)
- Improve chunking and metadata
- Compare semantic and hybrid retrieval
- Tune top-k and similarity thresholds; evaluate reranking
- Measure precision and recall
- Add out-of-scope and adversarial tests

Key learning: a useful agent needs more than a fluent model response. Retrieval evidence, honest uncertainty, validation, and a human-controlled action boundary are all part of the product.

THANK YOU • QUESTIONS & FEEDBACK